

Quiz 6 Reflection

SCIE 001 Physics

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February 2, 2026

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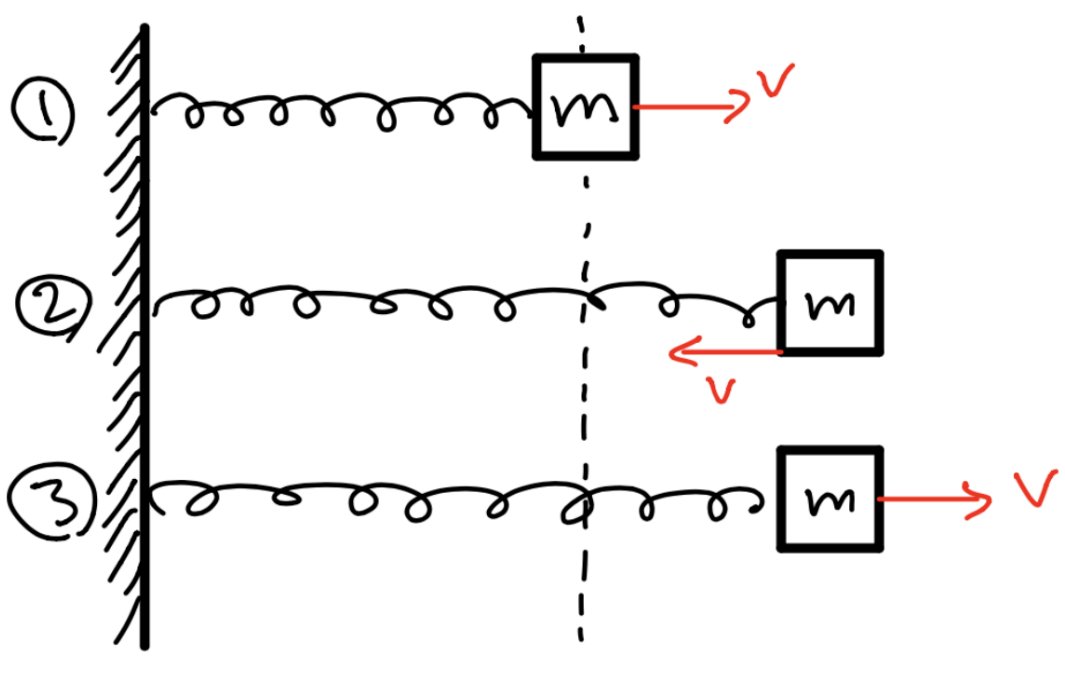
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1. Three Oscillators

1.1. Specify Question

Question 1: Three Oscillators

Consider 3 identical masses attached to 3 identical springs, each with identical speed at the moment shown below. The dashed line shows the equilibrium position and there is no damping. What can you say about the **maximum velocity** attained by each of these oscillators?



☐ $v_3 > v_1 > v_2$

☐ $v_1 > v_2 = v_3$

☐ $v_2 > v_1 > v_3$

☐ $v_2 > v_3 > v_1$

☐ $v_2 = v_3 > v_1$

☐ Not enough information to answer.

☐ $v_1 = v_2 = v_3$

☒ $v_3 > v_2 > v_1$

Correct answer

$v_2 = v_3 > v_1$

1.2. Diagnosis Phase

Maximum velocity is directly proportional to maximum amplitude.

$$v_{\text{max}} = A\omega$$

I misread the diagram and incorrectly thought that the 3rd mass-spring system was stretched further than the second spring in the snapshot.

If correct, the third mass would reach a greater amplitude than the second mass. I thought this because, when the second mass was traveling back outwards, it would be at the same position with opposite velocity, and since I mistook that the third was further outstretched with same velocity, the third would go farther - therefore, having a greater amplitude and greater maximum velocity.

Words: 93

1.3. Correction Phase

I was incorrect in reading that the blocks were at different distances when the snapshot was taken.

Reading the diagram properly, at some stretch x and some velocity v :

$$\begin{aligned} E_{\text{total}} &= E_{\text{kinetic}} + E_{\text{spring}} \\ E_{\text{max amplitude}} &= E_{\text{spring max amplitude}} + E_{\text{kinetic max amplitude}} \\ E_{\text{max amplitude}} &= E_{\text{total}} \quad \text{energy is conserved} \\ E_{\text{spring max amplitude}} + E_{\text{kinetic max amplitude}} &= E_{\text{kinetic}} + E_{\text{spring}} \\ \frac{1}{2}kA^2 + \frac{1}{2}m(0)^2 &= \frac{1}{2}mv^2 + \frac{1}{2}kx^2 \\ kA^2 &= mv^2 + kx^2 \\ A &= \sqrt{\frac{mv^2}{k} + x^2} \\ v_{\text{max}} &= \omega \sqrt{\frac{mv^2}{k} + x^2} \quad \text{since } v_{\text{max}} = \omega A \\ v_{\text{max}} &= \sqrt{\frac{k}{m}} \sqrt{\frac{mv^2}{k} + x^2} \quad \text{since } \omega = \sqrt{\frac{k}{m}} \\ v_{\text{max}} &= \sqrt{v^2 + \frac{kx^2}{m}} \end{aligned}$$

Now we can plug in what we know about the blocks to get expressions for their maximum velocities. Defining right as positive:

$$\begin{aligned} v_{\text{max } 1} &= \sqrt{v^2 + \frac{k(0)^2}{m}} = |v| \quad \text{since the first spring is at the equilibrium} \\ v_{\text{max } 2} &= \sqrt{(-v)^2 + \frac{k(x)^2}{m}} = \sqrt{v^2 + \frac{k(x)^2}{m}} \quad \text{where } x \text{ is the distance stretched} \\ v_{\text{max } 3} &= \sqrt{v^2 + \frac{k(x)^2}{m}} \\ |v| &< \sqrt{v^2 + \frac{k(x)^2}{m}} = \sqrt{v^2 + \frac{k(x)^2}{m}} \\ \therefore v_{\text{max } 1} &< v_{\text{max } 2} = v_{\text{max } 3} \\ \therefore v_2 &= v_3 > v_1 \end{aligned}$$

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